Candidate seat no:________________

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Fourth Semester of B.Tech. Theory (IT) Examination

December 2015

MA206 Probability, Statistics & Numerical Analysis

Date: 16/12/2015, Wednesday **Time: 01:30 p.m. to 04:30 p.m.** **Maximum Marks:70**

Instructions:

1. **This question paper comprises of two sections.**
2. **Section I and II must be attempted in separate answer sheets.**
3. **Make suitable assumptions and draw neat figures wherever if required.**
4. **Use of scientific non-programable calculator is allowed.**

SECTION I

Q.1. Write whether following statements are true or false. [07]

(a) If the mean of 5 values of variable x is 8, then $\sum_{i=1}^{5} x_i = 40$.

(b) The number of possible outcomes when $if \dots then \dots else$ statement executed four times is 8

(c) If A and B are two events on sample space S with probability function P(.), then $P(A \cup B') = P(A) + P(B) - P(A \cap B')$

(d) If a random variable X has Poisson probability distribution with mean =2, the variance of distribution is 2.

(e) The standard deviation of 6 values of variable is 10,therefore variance of variable is 36.

(f) The mode of data 5 3 2 4 5 is 5

(g) The value of Karl Pearson coefficient of correlation r satisfies $-1 \leq r \leq 1$.

Q.2. Answer the following questions. [14]

(a) An investigation on CPU performance gives the frequency distribution of attribute PRP (published relative performance in integers) [03]

Table 1: PRP range values:(Lower $<$ PRP $\leq$ Upper)

Lower	0	20	100	180	260	340	420	500
Upper	20	100	180	260	340	420	500	580
Number of Instances in Range	31	121	27	13	8	4	2	4

Represent the frequency distribution in Table 1 by Histogram.

(b) In a 2-week study of the productivity of workers, the following data were obtained on the total number of acceptable pieces which 100 workers produced: [04]

Table 2: Number of acceptable pieces

65	43	88	59	35	76	21	45	63	41	56	72	46	51	80	54	61	69	50	35
36	78	50	48	62	60	35	53	65	74	28	55	39	40	64	45	77	52	53	47
49	37	60	76	52	48	61	34	55	82	43	62	57	75	53	44	60	68	59	50
84	40	56	74	63	55	45	67	61	58	67	22	73	56	74	35	85	52	41	38
79	68	57	70	32	51	33	42	73	26	36	82	65	45	34	51	68	47	54	70

Group these figures (Table (2))into a distribution having the classes $20-29, 30-39, 40-49, \ldots 80-89$.

(c) A study with a group of 10 students of first-year undergraduate Engineering,carried out over a simulation environment named Deeds (Digital Electronics Education and Design Suite) which is used for e-learning in digital electronics. The environment provides learning materials through specialized browsers for the students, and asks them to solve various problems with different levels of difficulty.The table shows the final grades obtained by these students (Table 3). [07]

Table 3: Final grades of Students

Student ID	1	2	3	4	5	6	7	8	9	10
Marks(100)	94.5	44.0	30.0	38.5	78.0	8.5	18.5	60.0	40.5	64.0

Determine following summary measures of final grades of students
(i) Mean (ii) Median (iii) Standard deviation

OR

Q.2. Answer the following questions. [14]

(a) There are 18 computers in a store. Among them, 12 are brand new and 6 are refurbished. four computers are purchased for a student lab. From the first look, they are indistinguishable, so the four computers'are selected at random. Compute the probability that among the chosen computers, [03]
(i) none of them are refurbished. (ii) two are refurbished.

(b) Write mathematical properties of arithmetic mean. [04]

(c) A computer manager needs to know how efficiency of her new computer program depends on the size of incoming data. Efficiency will be measured by the number of pro- cessed requests per hour. Applying the program to data sets of different sizes, she gets the following results (Table 4), [07]

Table 4: Computer Program Efficiency data

Data size (gigabytes), x	6	7	7	8	10	10	15
Processed requests, y	40	55	50	41	17	26	16

(i) Construct the scatter plot of the data.

(ii) Determine a least-squares linear line through the data in table 4

Q.3. Answer the following questions. [14]

(a) Let there are 10 binary digits transmitted through communication channel and X denotes number of digits received correctly.If probability of successfully transmitting a digit is 0.8,Compute the probability that [05]

(i) none of digits transmitted correctly.

(ii) At least two digits transmitted correctly

Use Binomial distribution to answer your questions.

(b) Assume that the probability of k components malfunctioning within an interval of time t minutes in a system with a large number of components is given by the Poisson pmf with parameter $\lambda = 0.4t$.Compute the probabilities [05]

(i) No component malfunctioning in 10 minutes interval.

(ii) Exactly four components malfunctioning in 30 minutes interval.

(c) If X is a normal random variable with parameters $\mu = 3$ and $\sigma^2 = 9$, Compute probabilities(i) $P(X > 0)$ (ii) $P(2 < X < 5)$ [04]
You are given $P(Z \leq 1) = 0.841345, P(Z \leq 1/3) = 0.630559, P(Z \leq 2/3) = 0.747507$

OR

Q.3. Answer the following questions. [14]

(a) Define Uniform distribution over the interval $(a, b), a < b$. If a random variable X is uniformly distributed over (0,10), calculate the probability(i) $P\{X \leq 3\}$ (ii) $P\{X \geq 6\}$ (iii) $P\{3 \leq X \leq 8\}$ [07]

(b) Define Exponential distribution.Suppose the length of phone call (in minutes) is an exponential random variable with parameter $\lambda = \frac{1}{10}$ if someone immediately ahead of you at a telephone booth, find the probability that you will have to wait(i) more than 10 minutes (ii) between 10 and 20 minutes [07]

SECTION II

Q.4. Write whether following statements are true or false. [07]

(a) If $X = 10.2$ is true value and $\hat{X} = 10.5$ is its approximation,then error of $\hat{X}$ is -0.3.

(b) If $f(x)$ is real and continuous in the interval (a,b) and $f(a)f(b) < 0$ then there is atleast one real root between a and b.

(c) Using Newton Raphson method one can find the square root of a number $c > 0$,

(d) Let the curve $y = ae^{bx}$ to be fitted to the given data points $(x_i, y_i), i = 1, 2, \ldots, n$ to use least squares method to find a and b the one possible transformation is $\log_e y = \log_e a + bx$.

(e) Assume we have a table of values (x_i, y_i), $i = 0, 1, 2, 3$ of a function $y = f(x)$. One can obtain Lagrange interpolating polynomial of degree atmost three.

(f) Assume we have a table of values (x_i, y_i), $i = 0, 1, 2, 3$ of a function $y = f(x)$ and the values of x are equally spaced. Then relation $\Delta y_0 = \nabla y_1$ holds true.

Q.5. Answer the following questions. [14]

(a) Given the table of values(Table 5), Using Lagrange's interpolation evaluate $\sqrt{153}$ [07]

Table 5: Data for Interpolation

x	150	152	154	156
$y = \sqrt{x}$	12.247	12.329	12.410	12.489

(b) Determine a and b so that $y = ae^{bx}$ fits the data (Table 6) by the method of least-squares, Also compute y when $x = 7$ [07]

Table 6: Data for curve fit

x	2	4	6	8	10
y	4.077	11.084	30.128	81.897	222.62

OR

Q.5. Answer the following questions. [14]

(a) Determine the constants a and b by least-squares method such that $y = ab^x$, fits the data (Table 7). [07]

Table 7: Data for curve fit

x	2	4	6	8	10
y	3	13	32	57	91

(b) Consider the tabular values of function $f(x) = \cosh x$ (Table 8) [07]

Table 8: Data for Interpolation

x	0.5	0.6	0.7	0.8
$f(x) = \cosh x$	1.127626	1.185465	1.255169	1.337435

Compute $\cosh 0.56$ using appropriate Newton's interpolation formula.

Q.6. Answer the following questions. [14]

(a) Find a real root of the equation $x^3 - x - 1 = 0$ using Bisection method, in the interval [1,2]. Perform five iterations. [04]

(b) Apply the Runge-Kutta method of fourth order to the initial value problem $y' = x + y, y(0) = 0$ choosing $h = 0.2$ and compute $y(0.4)$ [04]

(c) Evaluate $I = \int\limits_0^1 \frac{1}{1+x^2} dx$ correct to three decimal places using trapezoidal rule with $h = 0.125$ [06]

OR

Q.6. Answer the following questions. [14]

(a) Apply the Euler's method to the initial value problem $y' = x + y, y(0) = 0$ choosing $h = 0.2$ and compute $y(1.0)$ [04]

(b) Determine the fourth root of 200 by finding the numerical solution of the equation $x^4 - 200 = 0$. Use Newton Raphson method. Start at $x = 8$ and carry out the first five iterations. [04]

(c) Use three point Gaussian integration formula to approximate $\int\limits_0^1 \frac{1}{1+x} dx$. Use following table values (9). [06]

Table 9: Weights for Gaussian Integration

n	$\pm u_i$	W_i
3	0.0	0.888889
	0.774597	0.555556